Decision-PGA and the Need for Decision-State Diagn

Zachary D. Michels, PhD

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A Prototype Vocabulary for Uncertainty Shape in Agentic AI Workflows

AI systems are moving from one-shot answer generation toward workflow participation. They retrieve evidence, extract data, route tasks, call tools, compare options, and recommend next actions. That shift changes the uncertainty problem. It is no longer enough to ask whether a model is confident in a final answer. A workflow often needs to know what kind of decision state the system is in before it acts.

Is the next action stable enough to proceed? Is the uncertainty concentrated between two plausible choices? Is it scattered across many alternatives? Is the system sensitive to a small boundary change? Has the decision state drifted over a multi-step trajectory?

Decision-PGA is a prototype method for studying those questions. It analyzes clouds of categorical probability vectors on the probability simplex using Fisher-Rao/square-root geometry, then describes the shape of the resulting uncertainty. The aim is not to replace task evaluation or human review. The aim is to make a decision state easier to inspect, compare, and route.

This article is a personal technical perspective, not an institutional statement. It uses no patient data, is not clinical validation, and does not describe a medical device or clinical decision support product.

The gap between confidence and action

Many AI tools already expose useful signals: confidence scores, token probabilities, calibrated probabilities, retrieval scores, agreement rates, human review flags, and task-specific benchmarks. Those signals matter. The gap appears when a system must decide what to do next.

Consider a workflow that can answer, ask a clarifying question, retrieve more evidence, route to a reviewer, abstain, or replan. Two cases can have similar entropy but require different responses. In one case, almost all uncertainty may lie between two options. In another, the probability mass may be scattered across many actions. A scalar score may say "uncertain" in both cases. A

- an elongated two-choice cloud suggests targeted clarification or comparison;
- a diffuse cloud suggests missing context or insufficient evidence;
- a boundary-sensitive cloud suggests assumptions or thresholds should be inspected;
- a drifting cloud suggests segmentation, replanning, or escalation.

The practical question is modest but important: can we build diagnostic tools that help AI workflows choose safer and more useful next actions under uncertainty?

Why healthcare is a useful application lens

Healthcare is not the only place this matters, but it is a useful lens because the workflows are high-accountability, document-heavy, and full of decisions where "low confidence" is too vague to be operationally helpful.

Examples worth studying include:

- message or request triage, where candidate actions might include answer, clarify, route, schedule, retrieve context, or escalate;
- policy and guideline retrieval, where sources may be relevant but incomplete, outdated, or in tension;
- medical document extraction, where uncertainty may reflect two plausible values, source-span ambiguity, table-row confusion, or OCR sensitivity;
- trial matching and eligibility review, where some criteria are clearly met, some are missing, and some conflict across sources;
- operational agents that coordinate multi-step work and may drift as they gather context.

These examples should be treated as research and evaluation targets, not as claims of deployment readiness. A diagnostic can describe the shape of a decision state; it does not determine clinical truth, prove safety, or replace domain review.

The broader public context supports caution. The FDA maintains information on AI-enabled medical devices and clinical decision support software:
https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-enabled-medical-devices
and
https://www.fda.gov/medical-devices/software-medical-device-samd/clinical-decision-support-software-frequently-asked-questions-faqs.
The ONC HTI-1 rule addresses transparency for predictive decision support interventions in certified health IT:
https://healthit.gov/regulations/hti-rules/hti-1-final-rule/. The WHO has
published guidance on ethics and governance of AI for health:

not. The point is that governance and evaluation need inspectable technical signals, and decision-state diagnostics may become one useful category of such signals.

What Decision-PGA does

Decision-PGA starts with repeated probability-like observations over candidate labels. The labels might be actions, extracted field values, evidence clusters, routing choices, or review outcomes. The current prototype then:

1. Normalizes probability vectors.
2. Maps them to the positive unit sphere with the square-root transform.
3. Computes an intrinsic mean.
4. Log-maps samples into a tangent space.
5. Computes a dispersion tensor and eigensystem.
6. Reports shape metrics such as total dispersion, PC1 fraction, anisotropy ratio, margin, label switching, and half-window geodesic drift.

That produces a compact diagnostic contract:

State	Possible workflow interpretation
Stable	Proceed if the task is in scope and other checks pass.
Binary ambiguity	Ask a targeted question or compare top candidates.
Diffuse uncertainty	Gather evidence or broaden context.
Boundary sensitive	Inspect assumptions, thresholds, or constraints.
Regime shift	Segment the task, replan, or escalate.

This contract is intentionally small. It is meant to be usable by software systems, not just by notebooks. A local command-line tool, report generator, or agent tool can call the diagnostic and receive a structured result.

Why geometry?

Probability vectors live on a simplex, not in ordinary unconstrained Euclidean space. Decision-PGA uses the square-root transform to place probability vectors on the positive unit sphere, where Fisher-Rao geometry becomes easier to work with. The result is a way to analyze dispersion directions, not only dispersion amount.

That matters because uncertainty can have shape. A cloud stretched along one direction is different from a cloud spread broadly across many choices. Two states may have similar entropy while suggesting different next actions. PGA metrics such as the first principal geodesic fraction and anisotropy ratio are attempts to capture that distinction.

agreement, calibration, and task-specific accuracy.

Application patterns worth testing

Tool and action selection

Agentic systems frequently choose among tools, routes, or next actions. A diagnostic could distinguish stable tool choice from two-tool ambiguity or diffuse uncertainty across the action set.

Retrieval and evidence conflict

Retrieval systems can surface sources that are relevant but not mutually consistent. A diagnostic over candidate evidence clusters or answer decisions could help decide whether to answer, retrieve more, cite uncertainty, or route to review.

Document extraction

Extraction systems often face ambiguous spans, conflicting values, incomplete tables, or uncertain entity associations. A decision-state diagnostic could separate a stable extracted value from a two-value dispute or a diffuse source-association problem.

Multi-step workflow monitoring

An agent may begin with one plan, gather new evidence, and gradually move into a different decision regime. Sliding-window diagnostics could help identify when a trajectory should be segmented, replanned, or escalated.

What must be proven next

The next step is not a bigger claim. It is better evidence. Decision-PGA needs tests that ask where it adds value over simpler metrics and where it is redundant.

Useful near-term tests include:

- entropy-matched examples where binary ambiguity and diffuse uncertainty should lead to different actions;
- fixture suites for tool selection, retrieval conflict, and document extraction;
- held-out synthetic scenarios with known decision states;
- comparisons against entropy, margin, agreement, drift, and switch-rate

public, non-patient fixtures. No clinical or operational claim should depend on private data, anecdote, or unreviewed workflow assumptions.

A practical open-source path

The healthiest way to publish this idea is to separate the concept from claims of maturity:

- publish the article and invite critique;
- keep examples synthetic or public;
- release the prototype only when its docs, license, tests, and limitations are clear;
- report negative or redundant findings alongside promising ones;
- treat healthcare examples as evaluation targets that require separate governance before real-world use.

Decision-PGA may turn out to be most useful as a small observability layer: a local diagnostic that helps an AI workflow decide whether to proceed, clarify, gather evidence, inspect sensitivity, segment, replan, or escalate.

Conclusion

As AI systems become more active participants in workflows, they will need ways to expose not only what they chose, but what kind of uncertainty surrounded the choice. Confidence scores are part of that story, but they are not the whole story.

Decision-PGA proposes a compact way to describe uncertainty shape over candidate decisions. It is early, model-neutral, and intentionally modest. The important claim is not that the method is ready for high-stakes deployment. The important claim is that decision-state diagnostics deserve to be made visible, tested, and improved before agentic AI systems become routine infrastructure.